



LevelUP: AI-Powered Career Operating System

Autonomous Multimodal Agent for Engineering Skill Mastery & Task Scheduling

Official Solution Documentation & Engineering Report

A Complete Multimodal Platform for Proactive Skill Gap Analysis, Voice-First Tutoring, and 1-Click Kanban Auto-Scheduling

Powered by MiniMax-M3 LLM / FastAPI / React 18 / Firebase Firestore / Web Speech API

Primary GitHub Repository:

github.com/Prajeeth-12/LevelUP-new

Secondary Remote:

github.com/Prajeeth-12/LevelUP_HCL

Active Branch:

new-theme-layrs

Live Application (Vercel):

level-up-new.vercel.app

Backend Service (Render):

levelup-new-backend.onrender.com

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Key Highlights: MiniMax-M3 Live LLM | Bidirectional Voice Tutor | 1-Click Kanban Auto-Scheduler | Vision & GitHub Inspector | Human-in-the-Loop Safety

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1 Executive Summary & Problem Statement

LevelUP is an AI-powered personal career operating system and autonomous learning companion engineered to bridge the modern software developer skill gap. Instead of overwhelming learners with static course catalogs or rigid checklists, LevelUP pairs each engineer with an active, multimodal AI mentor that dynamically benchmarks their skills against industry job descriptions, balances weekly study bandwidth, schedules Kanban sprints, and provides turn-by-turn conversational tutoring.

1.1 The Three Core Engineering Challenges

1. **Skill Overwhelm & Fragmentation:** Learners juggle scattered resources—tutorials, GitHub repositories, Kanban boards, and interview questions—without a centralized intelligence layer connecting them.
2. **Cognitive Fatigue & Backlog Clutter:** Traditional task boards accumulate hundreds of unprioritized items, leading to burnout rather than focused execution.
3. **Passive vs. Active Mastery:** Reading documentation without hands-on architecture verification or spoken interview practice creates an illusion of competence.

1.2 The LevelUP Architecture Solution

LevelUP unifies career roadmap synthesis, daily task execution, and real-time AI mentorship into a synchronized full-stack architecture:

- **MiniMax-M3 High-Context Intelligence:** Deep reasoning model capable of processing complex architectural prompts, user skills, and active roadmaps.
- **1-Click Kanban Smart Auto-Scheduler:** Dynamically organizes tasks into daily focus sprints based on user study bandwidth (e.g., 12h/week).
- **Voice-First 1-on-1 AI Tutor:** Spoken technical mock interviews and conceptual deep dives via Web Speech APIs.
- **Multimodal Vision & GitHub Inspector:** Inspects whiteboard diagrams, system schemas, and live GitHub repositories.
- **Human-in-the-Loop Safety Guardrails:** All AI-suggested task additions or skill updates require explicit user approval.

2 Authentication & User Onboarding

LevelUP provides enterprise-grade authentication powered by Firebase Authentication with Google OAuth and Email/Password flows. The UI is built with a warm minimalist editorial aesthetic.

Figure 1: LevelUP Authentication Suite (Sign In with Google OAuth & Password Flow)

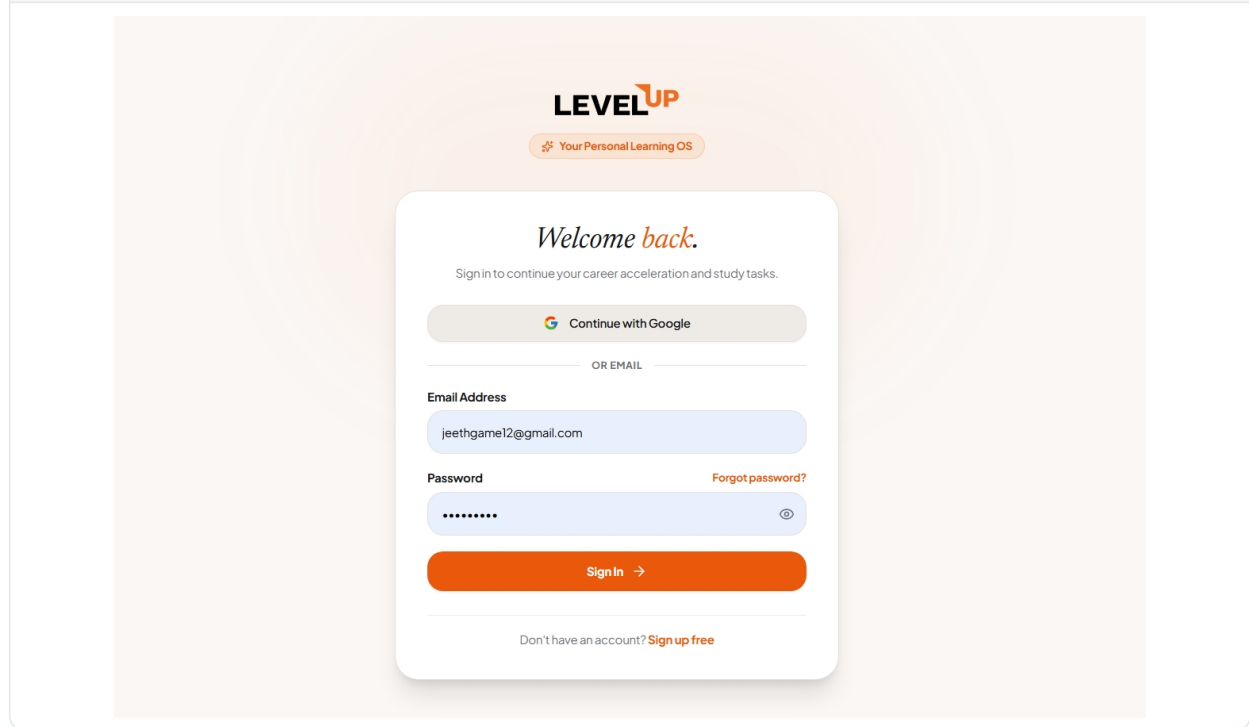


Figure 1: LevelUP Sign In Experience with Warm Layers Aesthetic, Google One-Tap, and Inline Password Recovery.

2.1 Security & User Experience Architecture

- **Google OAuth One-Tap:** Immediate onboarding with automatic Firestore profile initialization.
- **JWT Bearer Token Flow:** Client requests automatically retrieve fresh Firebase ID tokens and pass them to the FastAPI backend.
- **Guest Mode Fallback:** Allows exploration of roadmaps and features prior to authenticated state persistence.

3 Core Dashboard & Navigation Hub

The LevelUP dashboard serves as the central command center, aggregating active career roadmaps, daily prompts, cycle stats, and task queues in real time.

Figure 2: LevelUP Main Mission Control Dashboard

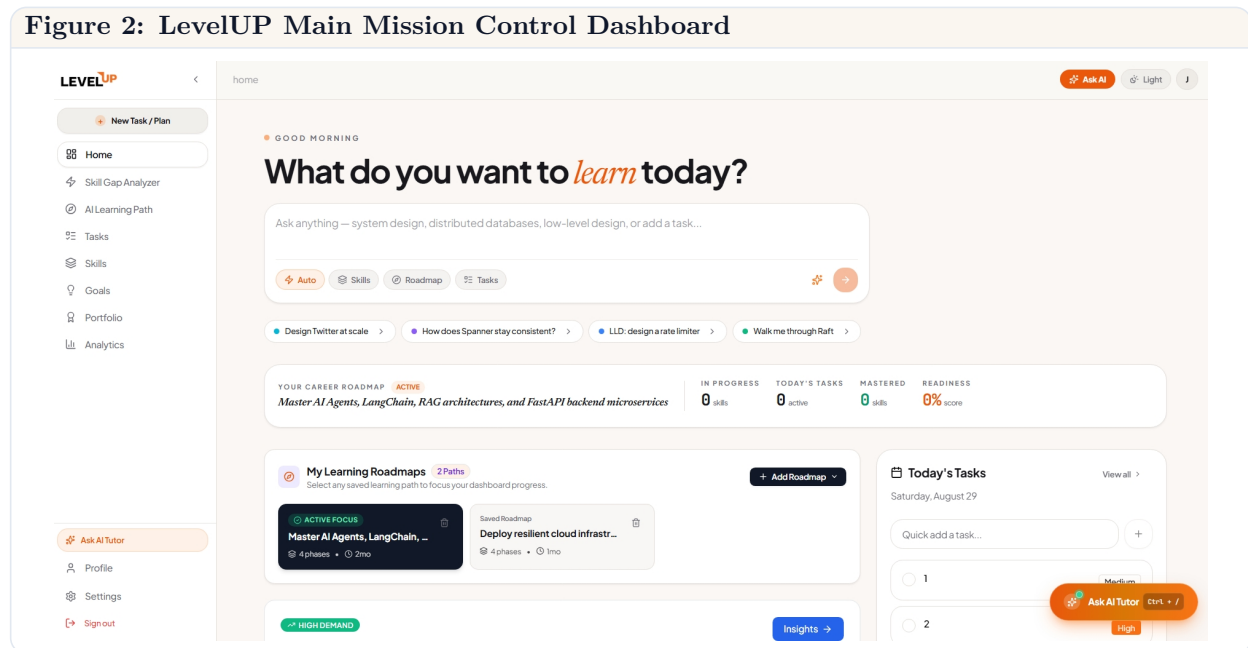


Figure 2: Mission Control Dashboard with Daily Question Prompt, Active AI Roadmaps, Cycle Stats, and Today's Tasks.

3.1 Dashboard Core Modules

- **Daily Knowledge Prompt:** Conversational input allowing learners to trigger technical deep dives (e.g., "Design Twitter at scale", "Walk me through Raft").
- **Cycle Stats Bar:** Live metrics displaying *In Progress Skills*, *Today's Tasks*, *Mastered Skills*, and the aggregate *Career Readiness Score*.
- **Active Roadmap Selector:** Multi-roadmap manager allowing instantaneous switching between personalized learning paths.
- **Static-Margin Rail Sidebar:** Fixed-position navigation that transitions smoothly without shifting main page content.

4 Multimodal AI Assistant & Voice Tutor

The LevelUP AI Drawer is an embedded, context-aware AI tutor powered by MiniMax-M3 LLM via an OpenAI-compatible API gateway.

Figure 3: LevelUP Multimodal AI Assistant Drawer with Voice Input & Context Awareness

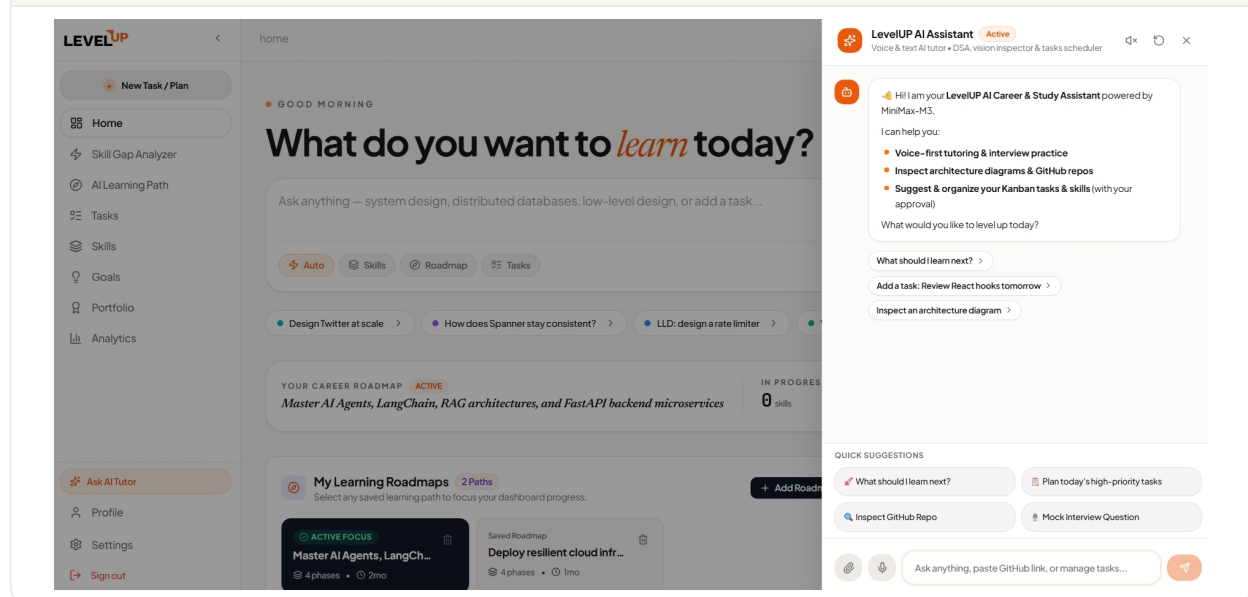


Figure 3: Interactive AI Chat Drawer featuring Voice Tutoring, File/Image Attachments, and Starter Prompts.

4.1 Key AI Superpowers

- **Voice-First 1-on-1 Tutoring:** Real-time Speech-to-Text input with animated waveform indicators and spoken Text-to-Speech audio playback.
- **Multimodal Vision Inspector:** Drag-and-drop support for system architecture diagrams, database schemas, and whiteboard photos.
- **GitHub Repository Scraping:** Inspects public GitHub repositories and diagnoses architectural gaps directly within the prompt context.
- **Human-in-the-Loop Action Cards:** AI suggestions are formatted as interactive cards with [+ Approve & Save] buttons.

5 3-Column Kanban Board & 1-Click Auto-Scheduler

The task management engine organizes execution into three columns: **To Do** (Backlog & Planned), **Current** (Active Focus & Today), and **Past** (Completed & History).

Figure 4: Tasks Kanban Board in Light Theme

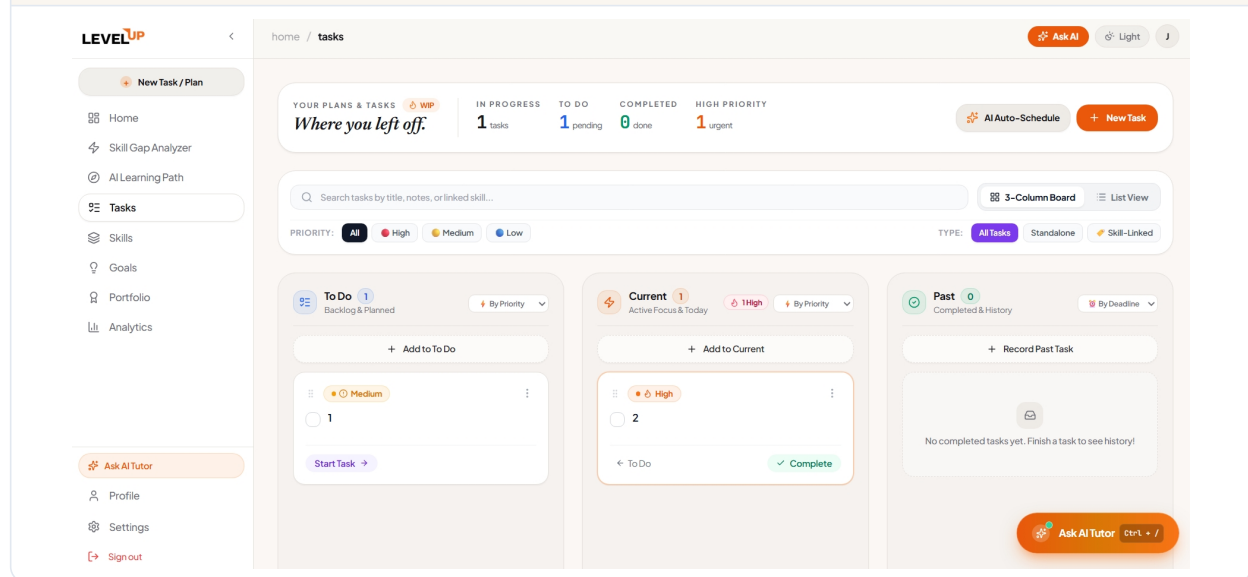


Figure 5: Tasks Kanban Board in Dark Theme

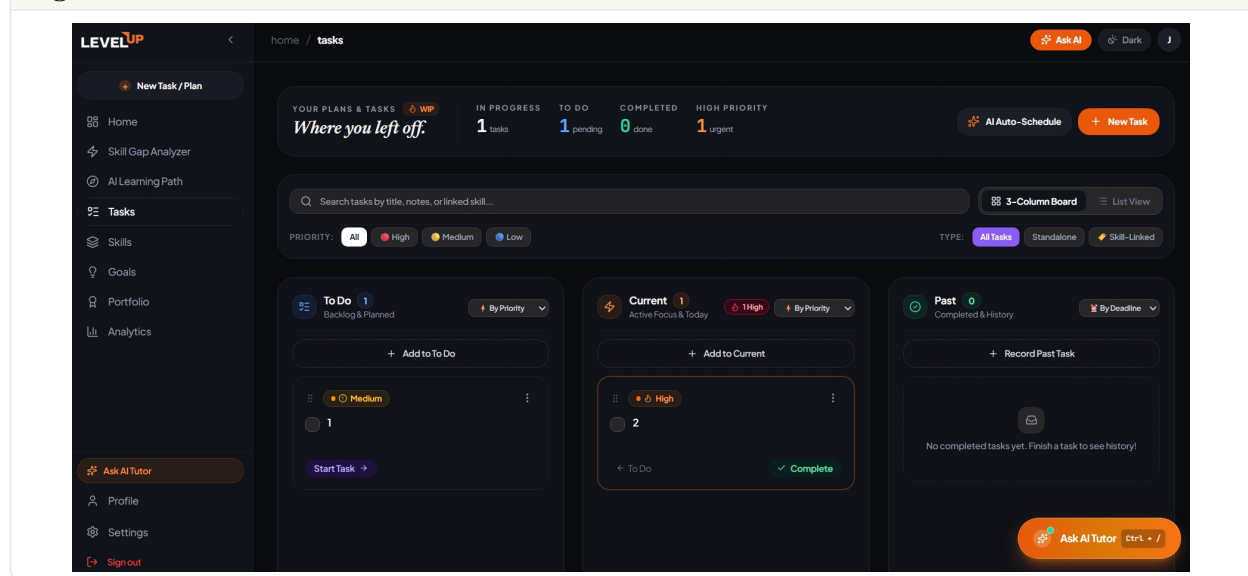


Figure 4: 3-Column Kanban Board in Light (top) and Dark (bottom) Themes with the AI Auto-Schedule System.

5.1 1-Click Kanban Smart Auto-Scheduler Algorithm

1. Evaluates open tasks against user-defined study bandwidth (e.g., 12 hours/week).
2. Identifies 2–4 high-leverage tasks for today's active sprint (2–4 hours).
3. Balances deadlines, cognitive difficulty, and skill dependencies.
4. Proposes structured sprint moves with pedagogical explanations before committing changes.

6 Skill Gap Analyzer & Job Description Matcher

The Skill Gap Analyzer benchmarks the learner's current competencies against target role requirements or pasted Job Descriptions.

Figure 6: Skill Gap Analyzer Intake & Matching Interface

Figure 5: Skill Gap Analyzer with Resume Upload, Study Bandwidth Slider, and Custom JD Parser.

6.1 Analyzer Capabilities

- **Dual-Mode Intake:** Choose from pre-configured engineering presets (Full Stack, Backend, DevOps, AI) or paste custom Job Descriptions.
- **Bandwidth Calibration:** Adjust weekly study bandwidth from 4h (casual) to 40h (bootcamp).
- **Automated Gap Synthesis:** Identifies missing core technologies, prerequisite gaps, and suggested learning sequences.

7 AI Learning Path Recommender

The Learning Path Recommender generates multi-phase, milestone-driven technical roadmaps tailored to specific career milestones.

Figure 7: Conversational AI Learning Path Recommender

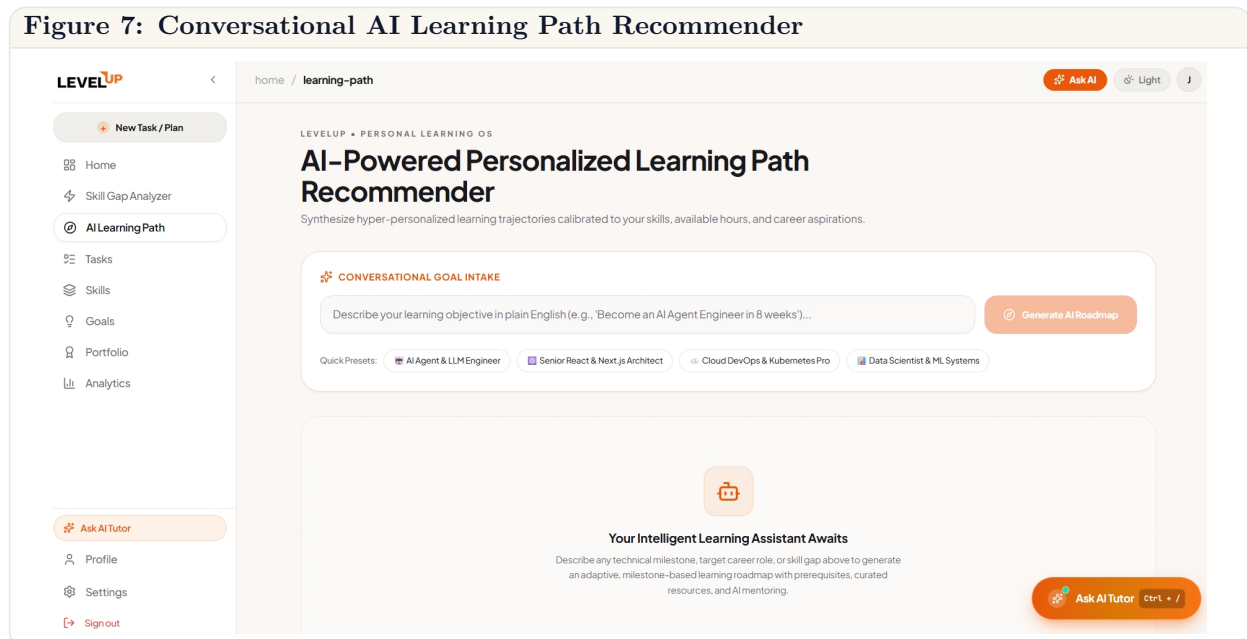


Figure 6: AI-Powered Personalized Learning Path Recommender with Conversational Goal Intake.

7.1 Roadmap Structure

- **Conversational Goal Intake:** Describe objectives in plain natural language (e.g., *"Become an AI Agent Engineer in 8 weeks"*).
- **Phased Milestones:** Generates structured learning phases with estimated durations, required subskills, and curated project assignments.
- **1-Click Adoption:** Instantly saves synthesized paths to Firestore and activates them across the main dashboard.

8 Career Portfolio & Readiness Verification

The Career Portfolio module provides verified evidence of engineering skills and aggregate career readiness.

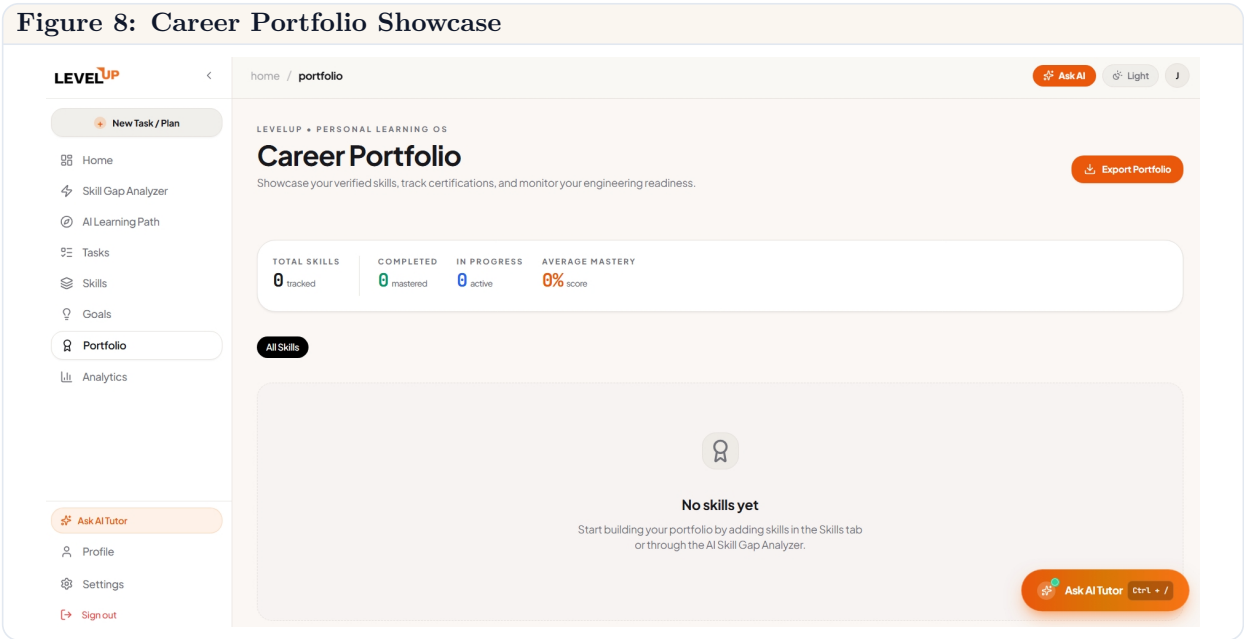


Figure 7: Career Portfolio Dashboard with Verified Skills, Completion Stats, and Export Portfolio Action.

8.1 Portfolio Features

- **Mastery Metrics:** Real-time calculation of completed skills, active skills, and average mastery percentages.
- **Exportable Portfolio:** One-click portfolio export for resume inclusion and hiring manager review.

9 Engineering Analytics & Study Insights

The Analytics engine aggregates task velocity, study consistency, and competency distribution metrics.

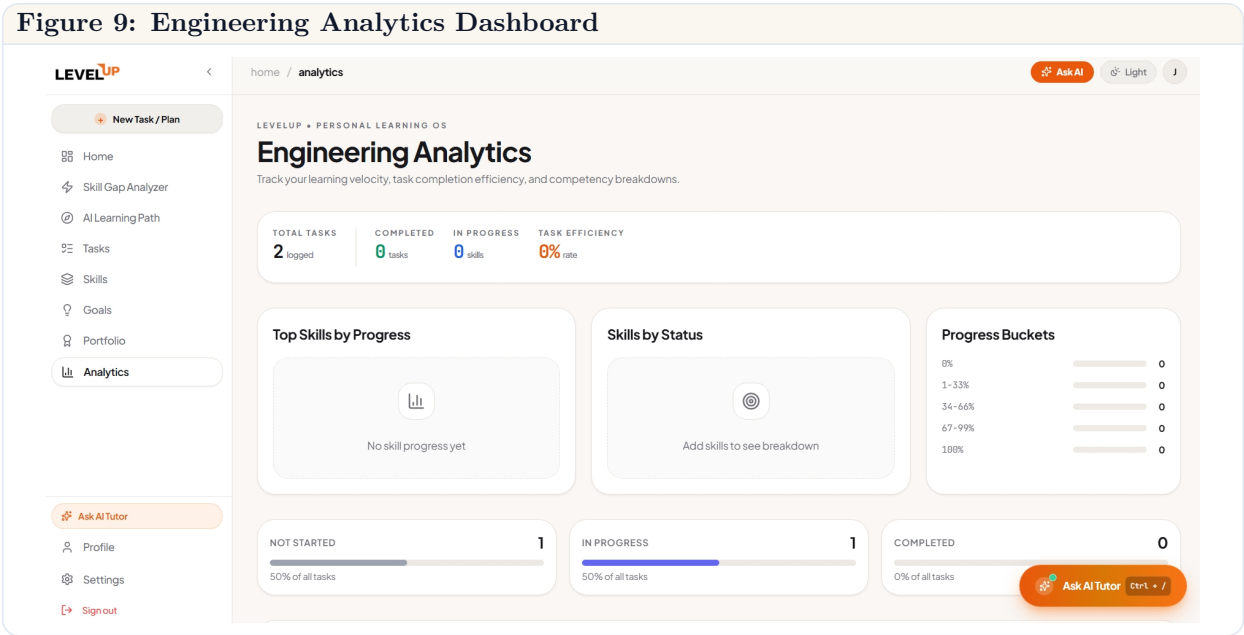


Figure 8: Engineering Analytics Dashboard with Task Efficiency, Progress Buckets, and Status Breakdowns.

9.1 Metrics Tracked

- **Task Efficiency Rate:** Ratio of completed sprint tasks to total scheduled backlog.
- **Progress Buckets:** Distribution of skills across 0%, 1–33%, 34–66%, 67–99%, and 100% mastery tiers.
- **Status Progress Bars:** Real-time visualization of Not Started vs. In Progress vs. Completed tasks.

10 User Profile, Settings & Data Portability

LevelUP provides full user autonomy over account information, technical preferences, and portable data backups.

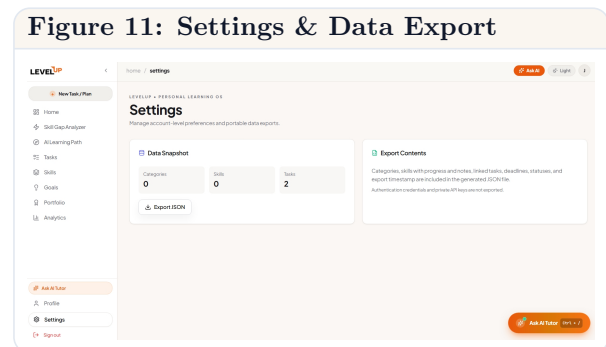
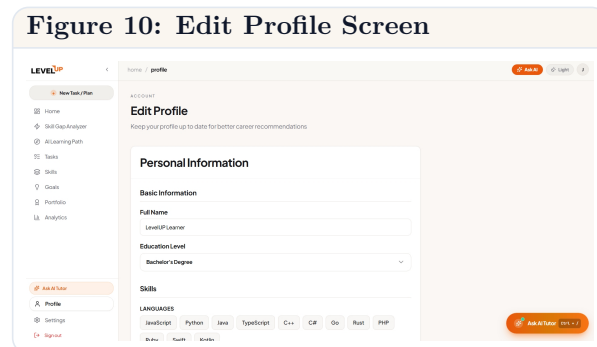


Figure 9: Left: User Profile editor with language and skill tags. Right: Settings with Portable JSON Data Export.

10.1 Data Autonomy & Portability

- **Profile Tagging:** Multi-language and framework competency tags for personalized AI recommendations.
- **1-Click JSON Export:** Complete export of categories, skills, tasks, deadlines, and notes for zero vendor lock-in.

11 Technical Architecture & API Reference

11.1 Full-Stack Architecture Matrix

Layer	Technology	Core Responsibilities
Frontend	React 18, Vite, Tailwind CSS, Framer Motion	Responsive Layrs theme, Kanban drag-and-drop, rich Mark-down rendering with code copying, audio visualizers.
Backend	FastAPI (Python 3.10+), Uvicorn	Asynchronous REST API, CORS middleware, JWT token decoding, and LLM prompt orchestration.
AI Engine	MiniMax-M3 LLM	High-context reasoning, multimodal vision analysis, and JSON action extraction.
Voice Layer	Web Speech Recognition & Synthesis	In-browser speech-to-text recording and turn-by-turn spoken audio playback.
Database	Google Cloud Firestore	Real-time NoSQL document store with reactive client snapshot listeners.
Auth	Firebase Authentication	JWT verification, Google OAuth, and secure guest access.

Table 1: LevelUP Full-Stack Architecture Matrix

11.2 Core Backend API Route Summary

Method	Endpoint	Functionality
POST	/ai/chat	Process chat messages with context, image inspection, and action generation.
POST	/ai/auto-organize-tasks	1-Click Kanban sprint scheduler based on weekly hour budget.
POST	/ai/skill-gap	Extract missing skills and generate learning paths from role/JD.
POST	/ai/generate-subskills	Decompose a skill into actionable subskills.
POST	/ai/task-plan	Prioritize and order tasks based on available study time.
POST	/ai/progress-insights	Generate coaching recommendations based on activity history.
POST	/api/students/profile	Create and update student career profiles in Firestore.
GET	/api/career/roadmap	Fetch active personalized career roadmap for a user.

Table 2: Core Backend API Route Summary

12 Local Setup, Execution & Deployment Guide

12.1 Step 1: Backend Setup (FastAPI)

1. Navigate to the backend directory: `cd backend`
2. Create and activate a virtual environment:
 - Windows: `python -m venv venv | .\venv\Scripts\activate`
 - macOS/Linux: `python3 -m venv venv | source venv/bin/activate`
3. Install dependencies: `pip install -r requirements.txt`
4. Configure backend/.env:
 - `MINIMAX_API_KEY=<your-key>`
 - `MINIMAX_BASE_URL=https://api.gmi-serving.com/v1`
 - `MINIMAX_MODEL=MiniMaxAI/MiniMax-M3`
5. Start the backend: `python -m uvicorn app.main:app -port 8000 -reload`

12.2 Step 2: Frontend Setup (React + Vite)

1. Navigate to frontend directory: `cd frontend`
2. Install dependencies: `npm install`
3. Start development server: `npm run dev`
4. Open in browser: <http://localhost:3000>

12.3 Step 3: Verified Production Deployments

- **Frontend Application (Vercel):** <https://level-up-new.vercel.app>
- **Backend Web Service (Render):** <https://levelup-new-backend.onrender.com>
- **GitHub Repository:** <https://github.com/Prajeeth-12/LevelUP-new> (Branch: `new-theme-layrs`)